

§ 3 静电场

§ 2.1 静电势及其微分方程

1. 静电势

$$\vec{E}(\vec{r}) = -\nabla \varphi(\vec{r})$$

$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V dv' \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} = \frac{1}{4\pi\epsilon_0} \int_V dv' \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$\oint d\vec{l} \cdot \vec{E} = \oint d\vec{l} \cdot (\nabla \times \vec{E}) = 0$$



$$\Rightarrow \int_{C1} d\vec{l} \cdot \vec{E} = \int_{C2} d\vec{l} \cdot \vec{E}$$

$$\text{可以定义 } \varphi(A) - \varphi(B) = \int_B^A d\vec{l} \cdot \nabla \varphi = - \int_B^A d\vec{l} \cdot \vec{E}$$

$$\text{规范: } \varphi(\infty) = 0$$

$$\varphi(P) = - \int_P^\infty d\vec{l} \cdot \nabla \varphi = \int_P^\infty d\vec{l} \cdot \vec{E}$$

2. 静电势的微分方程

$$\text{Poisson 方程: } \nabla^2 \varphi = \nabla \cdot \nabla \varphi = -\nabla \cdot \vec{E} = -\frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{D} = \epsilon \nabla \cdot \vec{E} = \rho_f$$

$$\Rightarrow \nabla^2 \varphi = -\frac{\rho_f}{\epsilon} = 0 \quad (\text{真空})$$

$$\text{Laplace 方程} \quad \nabla^2 \varphi = 0 \quad (\text{真空})$$

3. 电场极点的电荷密度

$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3}$$

$$\rho = -\epsilon_0 \nabla^2 \varphi = -\frac{1}{4\pi} \nabla^2 \frac{\vec{p} \cdot \vec{r}}{r^3}$$

为什么可以交换 $\nabla^2(\vec{p} \cdot \vec{r}) \rightarrow \vec{p} \cdot \nabla(\nabla^2 r)$

$$= \frac{1}{4\pi} \nabla^2 (\vec{p} \cdot \frac{1}{r}) = \frac{1}{4\pi} \vec{p} \cdot \nabla (\nabla^2 \frac{1}{r})$$

$$= \frac{1}{4\pi} \vec{p} \cdot \nabla [-4\pi \delta(\vec{r})] = -\vec{p} \cdot \nabla \delta(\vec{r})$$

4. 无限长电荷线元的静电势

$$\rho_{\lambda}(\vec{r}) = \begin{cases} \lambda & R \leq r \leq R+2a \\ 0 & \text{otherwise} \end{cases}$$

$$d\vec{p} = \rho_{\lambda} dz \vec{e}_z$$

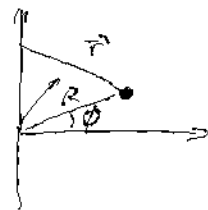
$$d\varphi = \frac{d\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^2} = \frac{\rho_{\lambda} dz}{4\pi\epsilon_0} \frac{1}{(R^2+z^2)^{3/2}} = \frac{\rho_{\lambda}}{4\pi\epsilon_0} \frac{R \cos\phi}{(R^2+z^2)^{3/2}} dz$$

$$\varphi = \int_{-\infty}^{\infty} d\varphi = \frac{\rho_{\lambda}}{4\pi\epsilon_0} R \cos\phi \int_{-\infty}^{\infty} \frac{dz}{(R^2+z^2)^{3/2}}$$

$$\int_{-\infty}^{\infty} \frac{dz}{(R^2+z^2)^{3/2}} = \frac{1}{R^2} \int_{-\infty}^{\infty} \frac{du}{(1+u^2)^{3/2}} \quad \text{令 } u = \tan\theta, \quad du = \frac{d\theta}{\cos^2\theta} = (1+u^2) d\theta = (1+u^2) du$$

$$= \frac{1}{R^2} \int_{-\pi/2}^{\pi/2} \cos\theta d\theta = \frac{2}{R^2}$$

$$= \frac{\rho_{\lambda} R \cos\phi}{4\pi\epsilon_0} \cdot 2 = \frac{1}{2\pi\epsilon_0} \frac{\rho_{\lambda}}{R} \cos\phi$$



5. Earnshaw定理

在无自由电荷分布的区域内, 静电势若不为常量, 则既没有极大也没有极小

证: 反证, 若有 $\varphi_0 = \varphi_{\max} / \varphi_{\min}$

$$\oint_{S_0} d\vec{S} \cdot \nabla\varphi > 0 \text{ 或 } < 0 \rightarrow \oint_{S_0} d\vec{S} \cdot \vec{E} < 0 \text{ 或 } > 0$$

与 $\rho=0$ 矛盾.

6. 介质界面的边值关系

$$(1) \varphi(r_1) = \varphi(r_2) = \int_{r_1}^{r_2} d\vec{r} \cdot \vec{E}$$

$$(2) (\vec{D}_2 - \vec{D}_1) \cdot \vec{n}_{21} = \sigma$$

$$(3) \epsilon_2 \frac{\partial\varphi_2}{\partial n_{21}} - \epsilon_1 \frac{\partial\varphi_1}{\partial n_{21}} = -\sigma$$

7. 金属导体表面静电势的边值关系

$$\begin{cases} \vec{n} \times \vec{E} = 0 \\ \vec{n} \cdot \vec{D} = \sigma \end{cases} \rightarrow \begin{cases} \varphi|_{\text{表面}} = \text{const.} \\ \epsilon_0 \frac{\partial\varphi}{\partial n} = -\sigma \end{cases}$$

$$\text{导体的总电荷: } Q = \oint_S \sigma dS = -\oint_S \epsilon_0 \frac{\partial\varphi}{\partial n} dS$$

例：计算静电状态下静电场作用在金属导体上的力。

外侧： $\epsilon_n = -\frac{\partial \varphi}{\partial n} = \frac{\sigma_f}{\epsilon_0} = \epsilon_{\text{面}} + \epsilon_{\text{其他}}$

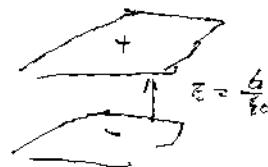
内侧： $\epsilon = \epsilon_{\text{面}} - \epsilon_{\text{其他}} = 0 \rightarrow \epsilon = \epsilon_{\text{面}} = \epsilon_{\text{其他}} = \frac{\sigma_f}{2\epsilon_0}$

$f = \epsilon \cdot \sigma_f^2 = \frac{\sigma_f^2}{2\epsilon_0}$

8. 电偶极子层静电势的边值关系。

$P_0(\vec{r}') = \sum_{a=1}^N q_a \delta(\vec{r}-\vec{r}') \cdot a$

$\varphi_2 - \varphi_1 = \frac{\Delta \cdot a}{\epsilon_0} = \frac{1}{\epsilon_0} P_0 \cdot \vec{n}_1$



9. 静电场的能量

$W = \frac{1}{4\pi\epsilon_0} \sum_{i,j} \frac{q_i q_j}{r_{ij}} = \frac{1}{4\pi\epsilon_0} \sum_{i,j} \frac{q_i q_j}{r_{ij}}$

$= \frac{1}{2} \sum_i q_i \sum_{j \neq i} \frac{1}{4\pi\epsilon_0} \frac{q_j}{r_{ij}} = \frac{1}{2} \sum_i q_i \varphi_i$

$= \frac{1}{2} \int \rho \varphi dV$

电磁场 $W = \frac{1}{2} \int_V \vec{E} \cdot \vec{D} dV \quad \vec{E} = -\nabla \varphi, \quad \nabla \cdot \vec{D} = \rho_f$

$= \frac{1}{2} \int_V -\nabla \varphi \cdot \vec{D} dV = \frac{1}{2} \int_V [\varphi (\nabla \cdot \vec{D}) - \nabla \cdot (\varphi \vec{D})] dV$

$= \frac{1}{2} \int_V \rho_f \varphi dV - \frac{1}{2} \oint_S \varphi (\vec{D} \cdot \vec{n}) dS = 0$

$= \frac{1}{2} \int_V \rho_f \varphi dV \quad V \rightarrow \infty, \quad \varphi \sim \frac{1}{r}, \quad D \sim \frac{1}{r^2}, \quad S \sim \frac{1}{r^2}.$

10. 相互作用能

空间中两电荷体系 ρ_1, ρ_2

$W = \frac{1}{2} \int_{V_1+V_2} (\varphi_1 + \varphi_2) (\rho_1 + \rho_2) dV$

$= \frac{1}{2} \int_{V_1} \varphi_1 \rho_1 dV + \frac{1}{2} \int_{V_2} \varphi_2 \rho_2 dV + \frac{1}{2} \int_{V_1} \varphi_2 \rho_1 dV + \frac{1}{2} \int_{V_2} \varphi_1 \rho_2 dV$

§ 3.2 静电势的多极展开

$$\varphi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{|\vec{r}-\vec{r}'|} dV' \quad \text{在远场的近似.}$$

1. 电多极展开

$$\begin{aligned} \frac{1}{|\vec{r}-\vec{r}'|} &= \frac{1}{r} - \vec{r}' \cdot \nabla \left(\frac{1}{r} \right) + \frac{1}{2!} (\vec{r}' \cdot \nabla)^2 \frac{1}{r} + \dots + \frac{(-1)^n}{n!} (\vec{r}' \cdot \nabla)^n \frac{1}{r} \\ &= \frac{1}{r} - R_i \frac{\partial}{\partial x_i} \left(\frac{1}{r} \right) + \frac{1}{2!} R_i R_j \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{1}{r} \right) + \dots \end{aligned}$$

$$\begin{aligned} \varphi(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \frac{1}{r} \int_V \rho(\vec{r}') dV' - \int_V \rho(\vec{r}') dV' R_i \frac{\partial}{\partial x_i} \left(\frac{1}{r} \right) + \frac{1}{2!} \int_V \rho(\vec{r}') dV' R_i R_j \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{1}{r} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} - \vec{p} \cdot \nabla \left(\frac{1}{r} \right) + \frac{1}{2!} D_{ij} \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{1}{r} \right) + \dots \right] \end{aligned}$$

$$\text{电荷极矩, } Q = \int_V \rho(\vec{r}') dV'$$

$$\text{电偶极矩, } \vec{p} = \int_V \rho(\vec{r}') dV' \vec{r}'$$

$$\text{电四极矩, } D_{ij} = \frac{1}{2} \int_V \rho(\vec{r}') dV' R_i R_j$$

$$\frac{\partial}{\partial x_i} \left(\frac{1}{r} \right) = -\frac{R_i}{r^3}$$

$$\frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{1}{r} \right) = \frac{3R_i R_j - \delta_{ij} r^2}{r^5}$$

$$\begin{aligned} \varphi(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} - \frac{\vec{p} \cdot \vec{r}}{r^3} + D_{ij} \frac{3R_i R_j - \delta_{ij} r^2}{r^5} + \dots \right) \\ &= \varphi^{(0)} + \varphi^{(1)} + \varphi^{(2)} \end{aligned}$$

3. 电四极矩.

$$\varphi^{(2)}(\vec{r}) = \frac{1}{4\pi\epsilon_0} D_{ij} \frac{3R_i R_j - \delta_{ij} r^2}{r^5}$$

$$Q=0, \vec{p}=0 \text{ 时.}$$

电四极矩不依赖于坐标原点的选取.

例: 构造如图带电体系, 求电势

$$\varphi = \frac{1}{4\pi\epsilon_0} \left[\frac{\vec{P} \cdot \vec{r}}{r^3} + \frac{\vec{P} \cdot (\vec{r} - s\vec{d})}{|\vec{r} - s\vec{d}|^3} \right]$$

$$= \frac{\vec{P}}{4\pi\epsilon_0} \cdot \left(-\frac{\vec{r}}{r^3} + \frac{\vec{r} - s\vec{d}}{|\vec{r} - s\vec{d}|^3} \right)$$

$$\approx \frac{\vec{P}}{4\pi\epsilon_0} \cdot \left\{ -\frac{\vec{r}}{r^3} + \left[\frac{1}{r^3} - s\vec{d} \cdot \nabla \left(\frac{1}{r^3} \right) \right] (\vec{r} - s\vec{d}) \right\}$$

$$\approx \frac{\vec{P}}{4\pi\epsilon_0} \cdot \left(-\frac{\vec{r}}{r^3} + \frac{\vec{r}}{r^3} - \frac{s\vec{d}}{r^3} - s\vec{d} \cdot \nabla \left(\frac{1}{r^3} \right) \vec{r} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \left[-\frac{\vec{P} \cdot \vec{d}}{r^3} - (\vec{P} \cdot \vec{r}) \vec{d} \cdot \nabla \left(\frac{1}{r^3} \right) \right] \quad \nabla \left(\frac{1}{r^3} \right) = -3 \frac{\vec{r}}{r^4} = -3 \frac{\vec{r}}{r^5}$$

$$= \frac{1}{4\pi\epsilon_0} \left(\frac{3(\vec{P} \cdot \vec{r})(\vec{d} \cdot \vec{r})}{r^5} - \frac{\vec{P} \cdot \vec{d}}{r^3} \right)$$

$$= \frac{1}{4\pi\epsilon_0} (3P_i r_i d_j r_j - r_i r_i P_j d_j) \frac{1}{r^5}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{1}{2} \left[3(P_i d_j + P_j d_i) r_i r_j - (P_i d_i + P_j d_j) r_i^2 \right] \frac{1}{r^5}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{1}{2} (P_i d_j + P_j d_i) \cdot \frac{3r_i r_j - \delta_{ij} r^2}{r^5}$$

$$= \frac{1}{4\pi\epsilon_0} D_{ij} \frac{3r_i r_j - \delta_{ij} r^2}{r^5} \quad D_{ij} = \frac{1}{2} (P_i d_j + d_i P_j)$$

$$D_{ij} = D_{ji}$$

φ 的化简过程:

对称二阶张量 D_{ij} 需要 6 个独立分量描述

$$\varphi^{(2)}(\vec{r}) = \frac{1}{\epsilon_0} \int_V \rho(\vec{r}') dV' R_i' R_j' \frac{3R_i R_j - \delta_{ij} R^2}{R^5}$$

$$R_i' R_j' \delta_{ij} R^2 = R'^2 R_i R_j \delta_{ij}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{1}{2} \int_V \rho(\vec{r}') dV' R_i' R_j' \frac{3R_i R_j}{R^5} - \frac{1}{2} \int_V \rho(\vec{r}') R'^2 dV' \frac{1}{R^3} R_i R_j \delta_{ij}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{1}{2} \int_V \rho(\vec{r}') (3R_i' R_j' - R'^2 \delta_{ij}) dV' \frac{R_i R_j}{R^5}$$

$$= \frac{1}{4\pi\epsilon_0} D_{ij} \frac{R_i R_j}{R^5}$$

D_{ij} 的化简过程:

$$= \frac{1}{2} \int_V \rho(\vec{r}') (3R_i' R_j' - R'^2 \delta_{ij}) dV'$$

$$\text{例: } \int_V \rho(\vec{r}') dV' = Q, \quad \int_V \rho(\vec{r}') R_i' dV' = 0$$

只需要 5 个独立分量描述.

例: 一均匀带电椭球, 椭球半径为 a, b, c . 总电荷量为 Q .

求带电椭球在远场的电势

$$x = R \sin \theta \cos \phi \quad y = R \sin \theta \sin \phi \quad z = R \cos \theta$$

$$x' = ar \sin \theta' \cos \phi' \quad y' = br \sin \theta' \sin \phi' \quad z' = cr \cos \theta'$$

$$dV' = dx' dy' dz' = abc r^2 \sin \theta' dr d\theta' d\phi'$$

$$\Phi_{11} = \frac{1}{2} \int_V \rho [3x^2 - (x^2 + y^2 + z^2)] abc r^2 \sin \theta' dr d\theta' d\phi'$$

$$= \frac{\rho abc}{2} \cdot \int_V (2x^2 - y^2 - z^2) r^2 dr \sin \theta' d\theta' d\phi'$$

$$= \frac{\rho abc}{2} \cdot \int_V [2a^2 r^2 \sin^2 \theta' \cos^2 \phi' - b^2 r^2 \sin^2 \theta' \sin^2 \phi' - c^2 r^2 \cos^2 \theta'] r^2 dr \sin \theta' d\theta' d\phi'$$

$$= \frac{\rho abc}{2} \int_V [2a^2 \sin^2 \theta' \cos^2 \phi' - b^2 \sin^2 \theta' \sin^2 \phi' - c^2 \cos^2 \theta'] r^5 dr \sin \theta' d\theta' d\phi'$$

$$= \frac{\rho abc}{2} \left[\frac{1}{3} r^5 \right]_0^R \cdot \int_0^\pi \sin \theta' d\theta' \int_0^{2\pi} [2a^2 \sin^2 \theta' \cos^2 \phi' - b^2 \sin^2 \theta' \sin^2 \phi' - c^2 \cos^2 \theta'] d\phi'$$

$$= \frac{\rho abc R}{10} \int_0^\pi \sin \theta' [2a^2 \sin^2 \theta' - b^2 \sin^2 \theta' - 2c^2 \cos^2 \theta'] d\theta'$$

$$= \frac{\pi \rho abc}{10} \left[\frac{2}{3} (2a^2 - b^2) - \frac{2}{3} \cdot 2c^2 \right] = \frac{\rho V}{10} (2a^2 - b^2 - c^2)$$

$$\Phi_{22} = \frac{1}{2} \int_V \rho [3y^2 - (x^2 + y^2 + z^2)] abc r^2 \sin \theta' dr d\theta' d\phi'$$

$$= \frac{\rho V}{10} (2b^2 - a^2 - c^2)$$

$$\Phi_{33} = \frac{\rho V}{10} (2c^2 - a^2 - b^2)$$

④ 非对称分量 = 0?

$$\varphi^{(2)}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_{R^3} \frac{\rho(\vec{r}')}{R^3} = \frac{1}{4\pi\epsilon_0} (\Phi_{11} x^2 + \Phi_{22} y^2 + \Phi_{33} z^2) \frac{1}{R^3}$$

$$= \frac{\rho V}{40\pi\epsilon_0 R^5} [(2a^2 - b^2 - c^2) x^2 + (2b^2 - a^2 - c^2) y^2 + (2c^2 - a^2 - b^2) z^2]$$

$$= \frac{\rho V}{40\pi\epsilon_0 R^5} [(2a^2 - b^2 - c^2) \sin^2 \theta \cos^2 \phi + (2b^2 - a^2 - c^2) \sin^2 \theta \sin^2 \phi + (2c^2 - a^2 - b^2) \cos^2 \theta]$$

$$\varphi(\vec{r}) = \frac{Q}{4\pi\epsilon_0 R} + \varphi^{(2)}(\vec{r})$$

5. 静电势在外场中的能量

$$W_{int} = \int_V \varphi_e(\vec{r}') \rho(\vec{r}') dV'$$

$$\varphi_e(\vec{r}') = \varphi_e(\vec{r}) + (\vec{r}' - \vec{r}) \cdot \nabla \varphi_e(\vec{r}) + \frac{1}{2} (\vec{r}' - \vec{r})^2 \cdot \nabla^2 \varphi_e(\vec{r}) + \dots$$

$$\begin{aligned} W_{int} &= \varphi_e(\vec{r}) \int_V \rho(\vec{r}') dV' + \int_V \rho(\vec{r}') (\vec{r}' - \vec{r}) dV' \cdot \nabla \varphi_e(\vec{r}) \\ &\quad + \frac{1}{2} \int_V \rho(\vec{r}') (\vec{r}' - \vec{r})^2 dV' \cdot \nabla^2 \varphi_e(\vec{r}) + \dots \\ &= Q \varphi_e(\vec{r}) + \vec{p} \cdot \nabla \varphi_e(\vec{r}) + \frac{1}{2} \vec{p} \cdot \nabla \nabla \varphi_e(\vec{r}) + \dots \\ &= W^{(0)} + W^{(1)} + W^{(2)} + \dots \end{aligned}$$

考虑 $W_{int} = W^{(1)}$ 的情况。

$$\vec{F} = -\nabla W^{(1)} = -\nabla (\vec{p} \cdot \nabla \varphi_e(\vec{r}))$$

$$= (\vec{p} \cdot \nabla) \vec{E}$$

$$C = -\frac{\partial W^{(1)}}{\partial \theta} = \frac{d(p E \cos \theta)}{d\theta} = -p E \sin \theta$$

$$= \vec{p} \times \vec{E}$$

6. 电偶极子的相互作用。

$$W_{int} = \vec{p}_2 \cdot \nabla \varphi_1(\vec{r}_2) = -\vec{p}_2 \cdot \vec{E}_1(\vec{r}_2)$$

$$\vec{E}_1(\vec{r}_2) = \frac{1}{4\pi\epsilon_0} \left[\frac{3\vec{p}_1 \cdot (\vec{r}_2 - \vec{r}_1)(\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^3} - \frac{\vec{p}_1}{|\vec{r}_2 - \vec{r}_1|^3} \right]$$

$$W_{int} = -\frac{1}{4\pi\epsilon_0} \frac{1}{|\vec{r}_2 - \vec{r}_1|^3} \left[\frac{3\vec{p}_1 \cdot (\vec{r}_2 - \vec{r}_1) \vec{p}_2 \cdot (\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^2} - \vec{p}_1 \cdot \vec{p}_2 \right]$$

考虑 $\vec{p}_1 \parallel \vec{p}_2$, 可用 $r(\theta) = |\vec{r}_2 - \vec{r}_1|$ 描述

$$W_{int} = \frac{p_1 p_2}{4\pi\epsilon_0 r^3} (1 - 3 \cos^2 \theta)$$

$$F = -\nabla W_{int}(r, \theta) = \frac{p_1 p_2}{4\pi\epsilon_0} \left[\frac{3 \cos^2 \theta - 1}{r^4}, -\frac{\theta \sin \theta \cos \theta}{r^4} \right]$$

7. 球坐标中的多极展开。

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§ 3.3 电场的唯一性定理

1. 在区域 V 内给定 ρ , 电荷分布的 E 且在 V 的边界面 S 上给定 $\varphi(R_3)$ 或 $\frac{\partial \varphi}{\partial n}$

则区域 V 内的电场唯一确定.

2.

证: 设区域 V 内电势为两解 φ, φ'' ,

$$\text{令 } \varphi = \varphi' - \varphi''$$

$$\begin{cases} \nabla^2 \varphi' = -\frac{\rho}{\epsilon_0} \\ \nabla^2 \varphi'' = -\frac{\rho}{\epsilon_0} \end{cases} \rightarrow \nabla^2 \varphi = 0$$

在 V 内任取 V_1, V_2 两边界面上:

$$\begin{cases} \varphi'_i = \varphi'_j \\ \epsilon_i \frac{\partial \varphi'_i}{\partial n} = \epsilon_j \frac{\partial \varphi'_j}{\partial n} \end{cases} \quad \begin{cases} \varphi''_i = \varphi''_j \\ \epsilon_i \frac{\partial \varphi''_i}{\partial n} = \epsilon_j \frac{\partial \varphi''_j}{\partial n} \end{cases} \rightarrow \begin{cases} \varphi_i = \varphi_j \\ \epsilon_i \frac{\partial \varphi_i}{\partial n} = \epsilon_j \frac{\partial \varphi_j}{\partial n} \end{cases}$$

① 在 V 的边界 S 上给定电势 $\varphi(R_3)$

$$\varphi|_S = \varphi'|_S - \varphi''|_S = 0$$

② 在 $\rightarrow$ 给定电势的法向导数 $\frac{\partial \varphi(R_3)}{\partial n}$

$$\epsilon \frac{\partial \varphi}{\partial n} \Big|_S = \epsilon \frac{\partial \varphi'}{\partial n} \Big|_S - \epsilon \frac{\partial \varphi''}{\partial n} \Big|_S = 0$$

$$\text{考虑 } \oint_{S_1} d\vec{S} \cdot \epsilon_1 \nabla \varphi \otimes \varphi = \int_{V_1} \nabla \cdot \nabla \varphi \cdot \epsilon_1 \varphi \otimes \varphi$$

$$= \int_{V_1} dV [\nabla(\epsilon_1 \varphi) \cdot \nabla \varphi + \epsilon_1 \varphi \nabla^2 \varphi]$$

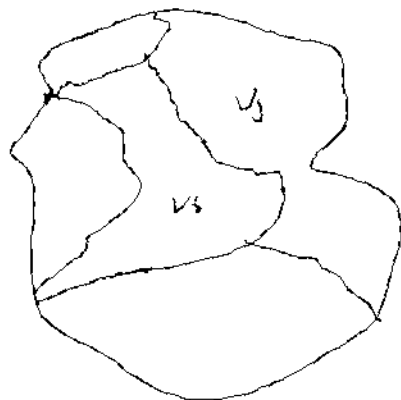
$$= \int_{V_1} dV \epsilon_1 [(\nabla \varphi)^2 + \varphi \nabla^2 \varphi] = \int_{V_1} dV \epsilon_1 |\nabla \varphi|^2$$

对 V 中所有子区域 V_i

$$\sum_i \oint_{S_i} d\vec{S} \cdot \epsilon_i \nabla \varphi \otimes \varphi = \left(\oint_S + \sum_i \oint_{S_{\text{inter}}} \right) d\vec{S} \cdot \epsilon_i \nabla \varphi \otimes \varphi$$

$$\oint_{S_{\text{inter}}} d\vec{S} \cdot \epsilon_i \nabla \varphi \otimes \varphi = \left(\int_{S_{i2}} d\vec{S}_{i2} \cdot \epsilon_i \nabla \varphi \otimes \varphi + \int_{S_{j1}} d\vec{S}_{j1} \cdot \epsilon_j \nabla \varphi \otimes \varphi \right)$$

$$= 0 \quad \varphi_i = \varphi_j, \quad \epsilon_i \frac{\partial \varphi_i}{\partial n} = \epsilon_j \frac{\partial \varphi_j}{\partial n}, \quad d\vec{S}_{j1} = -d\vec{S}_{i2}$$



$$\Rightarrow \int_S d\vec{S} \cdot \epsilon \nabla \phi = 0 = \int_V dV \epsilon |\nabla \phi|^2$$

$$\nabla \phi = 0 \quad \text{即} \quad \phi = \text{const.}$$

$$dS_{ji} \quad V_i \quad V_j$$

3. 金属导体存在时电势-性定理.

边界条件: 给定每个导体上的 Q_i / ϕ

(1) 第一类边值问题

给定 V' 中 ρ, ϵ 和 S 上 $\phi|_S, \left. \frac{\partial \phi}{\partial n} \right|_S$, 且每个导体电势 ϕ_i 给定.

则 V' 内电场唯一确定. 与一般情形-混合边界条件等价.

(2) 第二类边值问题.

给定 V' 中 ρ, ϵ 和 S 上 $\phi|_S, \left. \frac{\partial \phi}{\partial n} \right|_S$, 且每个导体带电量 Q_i 给定.

则 V' 内电场唯一确定.

证: 对于导体表面 S_i , 有边界条件

$$Q_i = \oint_{S_i} \epsilon \nabla \phi \cdot d\vec{S} = - \oint_{S_i} \epsilon \frac{\partial \phi}{\partial n} dS$$

反证法: 假设有两解 ϕ', ϕ'' ,

$$\text{令 } \phi = \phi' - \phi'' \rightarrow \nabla^2 \phi = 0$$

$$\begin{cases} \phi|_{S_i} = \phi'|_{S_i} - \phi''|_{S_i} \\ \epsilon \frac{\partial \phi}{\partial n} = \epsilon \frac{\partial \phi'}{\partial n} - \epsilon \frac{\partial \phi''}{\partial n} \end{cases} \quad \text{or}$$

$$\begin{cases} \oint_{S_i} \epsilon \frac{\partial \phi}{\partial n} = \oint_{S_i} \epsilon \left(\frac{\partial \phi'}{\partial n} - \frac{\partial \phi''}{\partial n} \right) = \frac{1}{\epsilon} (Q_i - Q_i) = 0 \\ \phi|_{S_i} = \phi'|_{S_i} - \phi''|_{S_i} = \text{const.} \end{cases}$$

$$\left[\int_S + \sum_i \oint_{S_i} \right] d\vec{S} \cdot (\nabla \phi \nabla \phi) = \sum_i \oint_{S_i} d\vec{S}_i \cdot \nabla \phi \nabla \phi$$

$$= - \sum_i \oint_{S_i} \phi|_{S_i} \oint_{S_i} \frac{\partial \phi}{\partial n} dS_i = 0$$

$$= \int_V \nabla \cdot (\epsilon \nabla \phi \nabla \phi) dV = \int_V [\epsilon |\nabla \phi|^2 + \epsilon \phi \nabla^2 \phi] dV = \epsilon \int_V |\nabla \phi|^2 dV$$

$$\Rightarrow \nabla \phi = 0, \quad \phi = \text{const.}$$

例: 不带电的金属球壳 S 内有一带电体 n , 证明

S 表面及以外的电场分布与 n 的形状, 位置无关.

证: 考虑 S 表面及以外空间, 其中没有电荷, 且 S 上 $\phi = \text{const.}$



边界条件: $V|_{\infty} = 0$, $\int_S \sigma dS = -Q_m$ 亦与 (r, θ) 位置无关.

例: 计算体系的电势分布及金属球壳上电荷分布.

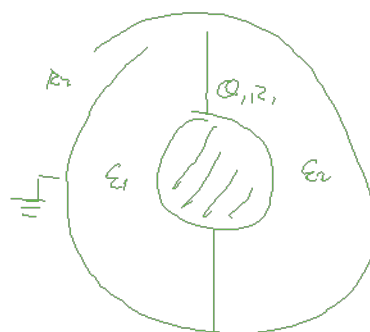
取试探解 $\varphi(r) = a + \frac{b}{r}$

外球壳, $a + \frac{b}{R_2} = 0 \rightarrow a = -\frac{b}{R_2}$, $\varphi = b(\frac{1}{r_2} - \frac{1}{r})$

内: $Q = \int -\epsilon \frac{\partial \varphi}{\partial n} dS = 2\pi R^2 (\epsilon_1 + \epsilon_2) \frac{b}{R_2}$

$\therefore b = \frac{Q}{2\pi(\epsilon_1 + \epsilon_2)}$, $\varphi = \frac{Q}{2\pi(\epsilon_1 + \epsilon_2)} (\frac{1}{r_2} - \frac{1}{r})$

外: $Q_{外} = \int -\epsilon \frac{\partial \varphi}{\partial n} dS = Q$?



§ 3.4 镜像法

1. 接地金属导体表面附近的点电荷.

$\varphi|_{z=0} = 0$, $\varphi|_{R \rightarrow \infty, z=0} = 0$ 根据边界条件 引入镜像电荷

$$\varphi(\vec{r}) = \varphi_a(\vec{r}) + \varphi_{\text{像}}(\vec{r})$$

其中 $\varphi_a(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + y^2 + (z-a)^2}}$

根据对称性: $\varphi_{\text{像}}(\vec{r}) = \varphi_{\text{像}}(\vec{r}) = -\frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + y^2 + (z+a)^2}}$

$$\varphi = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{x^2 + y^2 + (z-a)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z+a)^2}} \right]$$

$$\sigma_f = -\epsilon_0 \frac{\partial \varphi}{\partial n} \Big|_{z=0}$$

$$= -\frac{Q}{4\pi} \frac{\partial}{\partial z} \left[\frac{1}{\sqrt{x^2 + y^2 + (z-a)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z+a)^2}} \right] \Big|_{z=0}$$

$$= -\frac{Q}{4\pi} \left\{ -\frac{(z-a)}{[x^2 + y^2 + (z-a)^2]^{3/2}} + \frac{(z+a)}{[x^2 + y^2 + (z+a)^2]^{3/2}} \right\} \Big|_{z=0}$$

$$= -\frac{Qa}{2\pi} \frac{1}{(x^2 + y^2 + a^2)^{3/2}}$$

$$Q_{\text{像}} = \int_{S(z=0)} \sigma_f dS = -\frac{Qa}{2\pi} \iint \frac{1}{(x^2 + y^2 + a^2)^{3/2}} dx dy$$

$$= -\frac{Qa}{2\pi} \int_0^\infty r dr \int_0^{2\pi} \frac{1}{(r^2 + a^2)^{3/2}} d\theta$$

$$= -Qa \int_0^\infty \frac{r dr}{(r^2 + a^2)^{3/2}} \quad t = \frac{r}{a}$$

$$= -Q \int_0^\infty \frac{t dt}{(1+t^2)^{3/2}} = -Q (1+t^2)^{-1/2} \Big|_0^\infty = -Q$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2a)^2}$$

$$W_{\text{int}} = \int_\infty^F \vec{F} \cdot d\vec{l} = \frac{Q^2}{16\pi\epsilon_0} \int_\infty^a \frac{dr}{r^2} = -\frac{Q^2}{16\pi\epsilon_0 a}$$

2. 接地金属导体附近的点电偶极子

$$\vec{p} = p \cos \theta \vec{e}_y + p \sin \theta \vec{e}_z, \quad \vec{p}' = -p \cos \theta \vec{e}_y + p \sin \theta \vec{e}_z$$

$$\varphi = \varphi_p + \varphi_{p'} = \frac{1}{4\pi\epsilon_0} \left(\frac{\vec{p} \cdot \vec{R}}{R^3} + \frac{\vec{p}' \cdot \vec{R}'}{R'^3} \right)$$

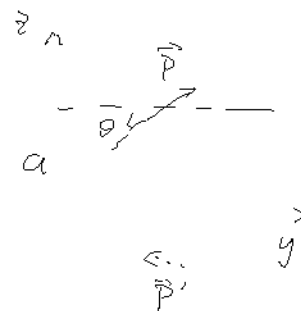
在 $\vec{p}$ 处,

$$\vec{E}_{p'} = \frac{1}{4\pi\epsilon_0} \left[\frac{3(\vec{p}' \cdot 2a\vec{e}_z) 2a\vec{e}_z}{(2a)^5} - \frac{\vec{p}'}{(2a)^3} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \frac{p}{8a^3} (\sin \theta \vec{e}_z - \sin \theta \vec{e}_z + \cos \theta \vec{e}_y)$$

$$= \frac{p \cos \theta}{32\pi\epsilon_0 a^3} \vec{e}_y$$

$$\vec{L} = \vec{p} \times \vec{E}_{p'} = \frac{p^2 \sin \theta \cos \theta}{32\pi\epsilon_0 a^3} \vec{e}_x$$



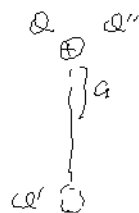
思考: 电偶极子受到作用力是向平面还是远离平面

3. 半无限大介质分界面附近的电荷

$$\varphi = \varphi_f + \varphi_{\text{OP}} + \varphi_{\text{面P}}$$

$$= \varphi_Q + \varphi_{Q'} = \frac{1}{4\pi\epsilon_2} \left(\frac{Q}{\sqrt{x^2+y^2+(z-a)^2}} + \frac{Q'}{\sqrt{x^2+y^2+(z+a)^2}} \right) \quad (z > 0) \quad \epsilon_2$$

$$= \varphi_{Q''} = \frac{1}{4\pi\epsilon_1} \frac{Q''}{\sqrt{x^2+y^2+(z-a)^2}} \quad (z < 0)$$



$$\text{边界条件} \begin{cases} \varphi_{z=0}|_{z=0} = \varphi_{z<0}|_{z=0} \\ \epsilon_2 \frac{\partial \varphi_{z=0}}{\partial n} \Big|_{z=0} = \epsilon_1 \frac{\partial \varphi_{z<0}}{\partial n} \Big|_{z=0} \end{cases}$$

$$\Rightarrow \begin{cases} \frac{1}{\epsilon_2} (Q + Q') = \frac{1}{\epsilon_1} Q'' & \text{由高斯定理} \\ Q - Q' = Q'' \end{cases} \quad \text{解得} \quad Q' = \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + \epsilon_1} Q, \quad Q'' = \frac{2\epsilon_1}{\epsilon_2 + \epsilon_1} Q$$

$$\varphi = \begin{cases} \frac{Q}{4\pi\epsilon_2} \left(\frac{1}{\sqrt{x^2+y^2+(z-a)^2}} + \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + \epsilon_1} \frac{1}{\sqrt{x^2+y^2+(z+a)^2}} \right) & (z > 0) \\ \frac{Q}{4\pi(\epsilon_1 + \epsilon_2)} \frac{1}{\sqrt{x^2+y^2+(z-a)^2}} & (z < 0) \end{cases}$$

$$Q \text{ 受电场力: } F = \frac{(\epsilon_2 - \epsilon_1) Q^2}{16\pi\epsilon_0 a^2 (\epsilon_2 + \epsilon_1)}$$

分界面上极化电荷:

$$\sigma_p = -(\vec{P}_2 - \vec{P}_1) \cdot \vec{e}_z$$

$$= -[\epsilon_2 - \epsilon_0] \vec{E}_2 - [\epsilon_1 - \epsilon_0] \vec{E}_1 \cdot \vec{e}_z \quad \vec{D}_1 = \epsilon_1 \vec{E}_1 = \vec{D}_2 = \epsilon_2 \vec{E}_2$$

$$= \epsilon_0 (\vec{E}_2 - \vec{E}_1) \cdot \vec{e}_z$$

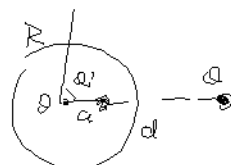
$$= \epsilon_0 \frac{1}{4\pi\epsilon_0 a^2} \left[-Q'' - (-Q - Q'') \right]$$

$$= \frac{Q}{4\pi a^2} \left(1 + \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + \epsilon_1} - \frac{2\epsilon_1}{\epsilon_2 - \epsilon_1} \right) = \frac{Q}{2\pi a^2} \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + \epsilon_1}$$

4. 金属导体球附近的点电荷.

$$\varphi = \varphi_a + \varphi_{a'}$$

$$\varphi_a = \frac{Q}{4\pi\epsilon_0} \frac{1}{|R^2 + d^2 - 2Rd\cos\theta|}, \quad \varphi_{a'} = \frac{Q'}{4\pi\epsilon_0} \frac{1}{|R^2 + a'^2 - 2Ra'\cos\theta|}$$



在球面上任一点:

$$\varphi|_{R=R_0} = \varphi_a|_{R=R_0} + \varphi_{a'}|_{R=R_0} = 0$$

$$\Rightarrow Q^2(R_0^2 + a^2 - 2R_0a\cos\theta) = Q'^2(R_0^2 + d^2 - 2R_0d\cos\theta)$$

$$\begin{cases} Q^2(R_0^2 + a^2) = Q'^2(R_0^2 + d^2) \\ Q^2a = Q'^2d \end{cases} \Rightarrow a = \frac{R_0^2}{d} \Rightarrow a = d \left(\frac{R_0}{d} \right)^2$$

$$Q' = -\frac{R_0}{d} Q$$

球外电势:

$$\varphi = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{|R^2 + d^2 - 2Rd\cos\theta|} - \frac{Q \cdot \frac{R_0}{d}}{|R^2 + a^2 - 2Ra\cos\theta|} \right)$$

$$\phi_f = -\epsilon_0 \frac{\partial \varphi}{\partial n} \Big|_{R=R_0}$$

$$= -\epsilon_0 \cdot \frac{\partial \varphi}{\partial R} \Big|_{R=R_0} = -\frac{Q}{4\pi} \frac{\partial}{\partial R} \left(\frac{1}{|R^2 + d^2 - 2Rd\cos\theta|} - \frac{\frac{R_0}{d}}{|R^2 + a^2 - 2Ra\cos\theta|} \right)$$

$$= -\frac{Q}{4\pi} \left[\left(-\frac{1}{2} \right) \frac{2R - 2d\cos\theta}{(R^2 + d^2 - 2Rd\cos\theta)^{3/2}} - \frac{\frac{R_0}{d}}{2} \left(-\frac{1}{2} \right) \frac{2R - 2a\cos\theta}{(R^2 + a^2 - 2Ra\cos\theta)^{3/2}} \right]$$

$$= \frac{Q}{4\pi} \left[\frac{R_0 - d\cos\theta}{(R_0^2 + d^2 - 2R_0d\cos\theta)^{3/2}} - \frac{R_0}{d} \frac{R_0 - a\cos\theta}{(R_0^2 + a^2 - 2R_0a\cos\theta)^{3/2}} \right]$$

$$= \frac{Q}{4\pi} (R_0^2 + d^2 - 2R_0d\cos\theta)^{-3/2} \left[(R_0 - d\cos\theta) - \frac{R_0}{d} \cdot \left(\frac{d}{R_0} \right)^3 (R_0 - \frac{R_0^2}{d} \cos\theta) \right]$$

$$= \frac{Q}{4\pi} (R_0^2 + d^2 - 2R_0d\cos\theta)^{-3/2} R_0 \left(1 - \frac{d^2}{R_0^2} \right)$$

$$Q' = \int \phi_f dS$$

$$= \int_0^\pi \frac{Q}{4\pi} (R_0^2 + d^2 - 2R_0d\cos\theta)^{-3/2} R_0 \left(1 - \frac{d^2}{R_0^2} \right) 2\pi R_0^2 \sin\theta d\theta$$

$$= -\infty$$

§ 3.5 Poisson 方程和 Green 函数法

Poisson 方程: $\nabla^2 \varphi(\vec{r}) = -\frac{1}{\epsilon_0} \rho(\vec{r})$, $\vec{r} \in V$

Green 函数: $\nabla^2 G(\vec{r}, \vec{r}') = -\frac{1}{\epsilon_0} \delta(\vec{r} - \vec{r}')$, $\vec{r}, \vec{r}' \in V$

在 $\vec{r}'$ 处放置单位点电荷对应的电势解.

Green 定理:

$$\varphi(\vec{r}), \psi(\vec{r}), \vec{r} \in V$$

$$\oint_S d\vec{S} \cdot (\psi \nabla \varphi) = \int_V dV \nabla \psi \cdot \nabla \varphi + \int_V dV \psi \nabla^2 \varphi$$

$$\oint_S d\vec{S} \cdot (\varphi \nabla \psi) = \int_V dV \nabla \varphi \cdot \nabla \psi + \int_V dV \varphi \nabla^2 \psi$$

$$\begin{aligned} \oint_V (\psi \nabla^2 \varphi - \varphi \nabla^2 \psi) dV &= \oint_S d\vec{S} \cdot (\psi \nabla \varphi - \varphi \nabla \psi) \\ &= \oint_S d\vec{S} \cdot \left(\psi \frac{\partial \varphi}{\partial n} - \varphi \frac{\partial \psi}{\partial n} \right) \end{aligned}$$

选取 $\varphi(\vec{r})$ 为待求电势解, $\psi(\vec{r})$ 为 Green 函数.

$$-\frac{1}{\epsilon_0} \int_V [G(\vec{r}, \vec{r}') \rho(\vec{r}) - \varphi(\vec{r}) \delta(\vec{r} - \vec{r}')] dV = \frac{1}{\epsilon_0} \left[\varphi(\vec{r}') - \int_V G(\vec{r}, \vec{r}') \rho(\vec{r}) dV \right]$$

$$\Rightarrow \varphi(\vec{r}') = \int_V G(\vec{r}, \vec{r}') \rho(\vec{r}) dV + \epsilon_0 \oint_S d\vec{S} \cdot \left[G(\vec{r}, \vec{r}') \frac{\partial \varphi}{\partial n} - \varphi \frac{\partial G}{\partial n} \right]$$

$\vec{r} \rightarrow \vec{r}'$

$$\varphi(\vec{r}) = \int_V G(\vec{r}, \vec{r}') \rho(\vec{r}') dV + \epsilon_0 \oint_S d\vec{S}' \cdot \left[G(\vec{r}, \vec{r}') \frac{\partial \varphi(\vec{r}')}{\partial n'} - \varphi(\vec{r}') \frac{\partial G(\vec{r}, \vec{r}')}{\partial n'} \right]$$

1. Dirichlet 条件: $G_n(\vec{r}, \vec{r}') = 0$ ($\vec{r}' \in S$, $\vec{r} \in V$)

$$\varphi(\vec{r}) = \int_V G_n(\vec{r}, \vec{r}') \rho(\vec{r}') dV - \epsilon_0 \oint_S d\vec{S}' \cdot \varphi(\vec{r}') \frac{\partial G_n(\vec{r}, \vec{r}')}{\partial n'}$$

只需解出 $G_n(\vec{r}, \vec{r}')$, 已知 $\varphi(\vec{r}') \ (\vec{r}' \in S)$, 所求出 $\varphi(\vec{r})$

2. Neumann 条件: $\frac{\partial G_n(\vec{r}, \vec{r}')}{\partial n'} = -\frac{1}{\rho \epsilon_0}$ ($\vec{r}' \in S$, $\vec{r} \in V$) A 为 S 的面积

$$\varphi(\vec{r}) = \int_V G_n(\vec{r}, \vec{r}') \rho(\vec{r}') dV + \epsilon_0 \oint_S d\vec{S}' \cdot G_n(\vec{r}, \vec{r}') \frac{\partial \varphi(\vec{r}')}{\partial n'} - \langle \varphi \rangle_S$$

$$\text{其中 } \langle \varphi \rangle_S = \frac{1}{A} \int_S \varphi(\vec{r}') dS'$$

只需解出 $G_N(\vec{r}, \vec{r}')$, 已知 $\frac{\partial \varphi}{\partial n}$, 可求出 $\varphi(\vec{r})$

3. 特殊情况

1) 无限空间.

$$G_D(\vec{r}, \vec{r}') = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{|\vec{r} - \vec{r}'|} + \left(-\frac{R_0}{R'}\right) \frac{1}{|\vec{r} - \frac{R_0^2}{R'^2} \vec{r}'|} \right]$$

$$\text{满足 } G_D(\infty, \vec{r}') = 0$$

2) 半无限空间

$$G_D(\vec{r}, \vec{r}') = \frac{1}{4\pi\epsilon_0} \left\{ \frac{1}{|\vec{r} - \vec{r}'|} + \frac{1}{|\vec{r} - \frac{R_0^2}{R'^2} \vec{r}'|} \right\}$$

3) 球外空间

$$G_D(\vec{r}, \vec{r}') = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{|\vec{r} - \vec{r}'|} + \left(-\frac{R_0}{R'}\right) \frac{1}{|\vec{r} - \frac{R_0^2}{R'^2} \vec{r}'|} \right]$$

4) 平面上半球面电荷空间.

$$\text{平面: } Q \rightarrow Q, Q'$$

$$\text{半球: } Q, Q' \rightarrow Q, Q', Q'', Q'''$$

$$\vec{r}'' = (x', y', -z')$$

$$G_D(\vec{r}, \vec{r}') = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{|\vec{r} - \vec{r}'|} + \left(-\frac{R_0}{R'}\right) \frac{1}{|\vec{r} - \frac{R_0^2}{R'^2} \vec{r}'|} - \frac{1}{|\vec{r} - \vec{r}''|} + \left(-\frac{R_0}{R''}\right) \frac{1}{|\vec{r} - \frac{R_0^2}{R''^2} \vec{r}''|} \right]$$

$$\begin{aligned} &= \frac{1}{4\pi\epsilon_0} \left[\left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-1/2} - \left[(x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-1/2} \right. \\ &\quad \left. - \frac{R_0}{R'} \left[\left(x - \frac{R_0^2}{R'^2} x' \right)^2 + \left(y - \frac{R_0^2}{R'^2} y' \right)^2 + \left(z - \frac{R_0^2}{R'} z' \right)^2 \right]^{-1/2} \right. \\ &\quad \left. + \frac{R_0}{R'} \left[\left(x - \frac{R_0^2}{R'^2} x' \right)^2 + \left(y - \frac{R_0^2}{R'^2} y' \right)^2 + \left(z + \frac{R_0^2}{R'} z' \right)^2 \right]^{-1/2} \right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{4\pi\epsilon_0} \left[(R^2 + R'^2 - 2RR' \cos \alpha_1)^{-1/2} - (R^2 + R'^2 - 2RR' \cos \alpha_2)^{-1/2} \right. \\ &\quad \left. - \left(\frac{R'^2 R^2}{R_0^2} + R_0^2 - 2RR' \cos \alpha_1 \right)^{-1/2} + \left(\frac{R'^2 R^2}{R_0^2} + R_0^2 - 2RR' \cos \alpha_2 \right)^{-1/2} \right] \end{aligned}$$

$$\cos \alpha_1 = \sin \theta \sin \theta' \cos(\phi - \phi') + \cos \theta \cos \theta'$$

$$\cos \alpha = \sin \theta \sin \theta' \cos(\phi - \phi') - \cos \theta \cos \theta'$$

例：一接地无限大金属导体平面上有一凸起的金属导体半球，其与导体平面相互绝缘。

半球半径为 R_0 ，与电势为 φ_0 电源相连，试求导体平面外各处电势分布。

$$\rho(\vec{r}') = 0$$

$$\varphi(\vec{r}) = -\epsilon_0 \oint_S \varphi(\vec{r}') \frac{\partial G(\vec{r}, \vec{r}')}{\partial n'} dS', \quad \frac{\partial G(\vec{r}, \vec{r}')}{\partial n'} = - \frac{\partial G(\vec{r}, \vec{r}')}{\partial R'} \Big|_{R=R_0}$$

$$\frac{\partial}{\partial R'} G(\vec{r}, \vec{r}') \Big|_{R=R_0}$$

$$= \frac{1}{4\pi\epsilon_0} \left[(R^2 + R'^2 - 2RR' \cos \alpha_1)^{-1/2} + (R^2 + R'^2 - 2RR' \cos \alpha_2)^{-1/2} \right. \\ \left. - \left(\frac{R'^2 R^2}{R_0^2} + R_0^2 - 2RR' \cos \alpha_1 \right)^{-1/2} + \left(\frac{R'^2 R^2}{R_0^2} + R_0^2 - 2RR' \cos \alpha_2 \right)^{-1/2} \right] \Big|_{R=R_0}$$

$$= \frac{1}{4\pi\epsilon_0} \left(-\frac{1}{2} \right) \left[(R^2 + R'^2 - 2RR' \cos \alpha_1)^{-3/2} [2R' - 2R \cos \alpha_1] \right. \\ - (R^2 + R'^2 - 2RR' \cos \alpha_2)^{-3/2} [2R' - 2R \cos \alpha_2] \\ - \left(\frac{R'^2 R^2}{R_0^2} + R_0^2 - 2RR' \cos \alpha_1 \right)^{-3/2} \left(\frac{2R'R^2}{R_0^2} - 2R \cos \alpha_1 \right) \\ \left. + \left(\frac{R'^2 R^2}{R_0^2} + R_0^2 - 2RR' \cos \alpha_2 \right)^{-3/2} \left(\frac{2R'R^2}{R_0^2} - 2R \cos \alpha_2 \right) \right] \Big|_{R=R_0}$$

$$= -\frac{1}{8\pi\epsilon_0} \left[(R^2 + R_0^2 - 2RR_0 \cos \alpha_1)^{-3/2} (2R_0 - 2R \cos \alpha_1 - 2\frac{R^2}{R_0} + 2R \cos \alpha_1) \right. \\ \left. - (R^2 + R_0^2 - 2RR_0 \cos \alpha_2)^{-3/2} (2R_0 - 2R \cos \alpha_2 - 2\frac{R^2}{R_0} + 2R \cos \alpha_2) \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[(R^2 + R_0^2 - 2RR_0 \cos \alpha_1)^{-3/2} - (R^2 + R_0^2 - 2RR_0 \cos \alpha_2)^{-3/2} \right] \left(\frac{R^2}{R_0} - R_0 \right)$$

$R \gg R_0$

$$\approx \frac{1}{4\pi\epsilon_0} \frac{1}{R^3} (1 + 3\frac{R_0}{R} \cos \alpha_1 - 1 - 3\frac{R_0}{R} \cos \alpha_2) \frac{R^2}{R_0}$$

$$= \frac{3}{4\pi\epsilon_0} \frac{1}{R^3} \frac{R_0}{R} (\cos \alpha_1 - \cos \alpha_2) \frac{R^2}{R_0} = \frac{3(\cos \alpha_1 - \cos \alpha_2)}{4\pi\epsilon_0 R^2}$$

$$\varphi(\vec{r}) = \epsilon_0 \oint_S d\vec{S}' \cdot \nabla \varphi = \frac{3(\cos \alpha_1 - \cos \alpha_2)}{4\pi\epsilon_0 R^2}$$

$$= \epsilon_0 \varphi_0 \frac{R_0^2}{R^2} \cdot \frac{3}{4\pi R^2} \int_0^{\frac{\pi}{2}} d\theta' \int_0^{2\pi} d\phi' \sin \theta' 2\cos \theta \cos \theta'$$

$$= \frac{3\varphi_0 \cos\theta R^2}{4\pi R^2} 2\pi \int_0^{\pi/2} 2 \sin\theta' \cos\theta' d\theta'$$

$$= \frac{3}{2} \varphi_0 \frac{\cos\theta \cdot R^2}{R^2} \sim \frac{\vec{p} \cdot \vec{R}}{4\pi\epsilon_0 R}$$

§ 3.6 Laplace 方程分离变量法.

$$\nabla^2 \varphi = 0$$

$$\varphi(u, v, w) = A_\alpha(u) B_\beta(v) C_\gamma(w)$$

1. 笛卡尔对称边界

$$\nabla^2 \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

$$\varphi(x, y, z) = X(x) Y(y) Z(z)$$

$$\hookrightarrow \frac{1}{x} \frac{d^2 x}{dx^2} + \frac{1}{y} \frac{d^2 y}{dy^2} + \frac{1}{z} \frac{d^2 z}{dz^2} = 0$$

$$\text{设 } \frac{1}{x} \frac{d^2 x}{dx^2} = k_x^2, \quad \frac{1}{y} \frac{d^2 y}{dy^2} = k_y^2, \quad \frac{1}{z} \frac{d^2 z}{dz^2} = k_z^2$$

$$\text{则 } k_x^2 + k_y^2 + k_z^2 = 0$$

$$\text{解得 } X(x) = \begin{cases} A_0 + B_0 x & (k_x = 0) \\ A_1 e^{k_x x} + B_1 e^{-k_x x} & (k_x \neq 0) \end{cases}$$

2. 球对称边界

$$\nabla^2 \varphi = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial \varphi}{\partial R} \right) + \frac{1}{R^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \varphi}{\partial \theta} \right) + \frac{1}{R^2 \sin^2\theta} \frac{\partial^2 \varphi}{\partial \phi^2} = 0$$

$$\varphi = \frac{u(R)}{R} P(\theta) Q(\phi)$$

$$\frac{1}{u} \frac{d^2 u}{dR^2} + \frac{1}{R^2 P \sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{dP}{d\theta} \right) + \frac{1}{R^2 Q \sin^2\theta} \frac{d^2 Q}{d\phi^2} = 0$$

$$R^2 \sin^2\theta \left[\frac{1}{u} \frac{d^2 u}{dR^2} + \frac{1}{R^2 P \sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{dP}{d\theta} \right) \right] + \frac{1}{Q} \frac{d^2 Q}{d\phi^2} = 0$$

$$\hookrightarrow \frac{1}{Q} \frac{d^2 Q}{d\phi^2} = -m^2, \text{ 解得 } Q = e^{\pm im\phi}$$

为 $\alpha(\psi + \pi/2) = \alpha(\psi)$, m 必须取偶数.

$$\frac{R^2}{u} \frac{d^2 u}{dR^2} + \left[\frac{1}{P \sin \theta} \frac{d}{d\theta} (\sin \theta \frac{dP}{d\theta}) - \frac{m^2}{\sin^2 \theta} \right] = 0$$

$$\Rightarrow \begin{cases} \frac{R^2}{u} \frac{d^2 u}{dR^2} = C \approx (l+1) & \text{由 } \square \text{ 解的性质规定,} \\ \frac{1}{\sin \theta} \frac{d}{d\theta} (\sin \theta \frac{dP}{d\theta}) + (C - \frac{m^2}{\sin^2 \theta}) P = 0 \end{cases}$$

解得 $u(R)/R = AR^l + B \frac{1}{R^{l+1}}$

$P(\theta) = P_l^m(\theta)$ 为连带 Legendre 多项式.

3. 轴对称的球形边界.

此时 $m=0$, $P_l^m(\theta)$ 退化为 Legendre 多项式.

$$\begin{aligned} \varphi(R, \theta) &= \frac{u(R)}{R} P(\theta) \\ &= \sum_l (A_l R^l + B_l \frac{1}{R^{l+1}}) P_l(\cos \theta) \end{aligned}$$

例: 由上、下半球组成的空心球壳, 上、下半球壳的电势分别为 $+V, -V$,

球壳半径为 a , 求球壳内、外的电势分布.

$$\varphi = \begin{cases} \sum_l A_l R^l P_l(\cos \theta) & (R < a) \\ \sum_l B_l \frac{1}{R^{l+1}} P_l(\cos \theta) & (R > a) \end{cases}$$

$$\varphi(a, \theta) = \begin{cases} +V & (\theta < \frac{\pi}{2}) \\ -V & (\theta > \frac{\pi}{2}) \end{cases}$$

$$A_l = \frac{1}{a^l} \cdot \frac{2l+1}{2} \int_0^\pi \varphi(a, \theta) P_l(\cos \theta) \sin \theta d\theta$$

$$= \frac{2l+1}{2a^l} V \left(\int_0^{\frac{\pi}{2}} - \int_{\frac{\pi}{2}}^\pi \right) P_l(\cos \theta) \sin \theta d\theta \quad \int_{-1}^1 dx P_l(x) P_m(x) = \frac{2}{2l+1} \delta_{lm}$$

$$B_l = \frac{(2l+1)a^{l+1}}{2} \left(\int_0^{\frac{\pi}{2}} - \int_{\frac{\pi}{2}}^\pi \right) P_l(\cos \theta) \sin \theta d\theta$$

2. 将半径为 R_0 , 介电常数为 ϵ_s 的带电颗粒放置于均匀外场 $\vec{E}_0$ 中,

颗粒周围为真空, 求颗粒内、外的电势分布.

$$\varphi_{\text{内}} = \sum_l A_l R^l P_l(\cos \theta) \quad (R < R_0)$$

$$\varphi_{\text{外}} = \sum_l (C_l R^l + \frac{D_l}{R^{l+1}}) P_l(\cos \theta) \quad (R > R_0)$$

$$R \rightarrow \infty, \quad \vec{E} = \vec{E}_0 \cdot \vec{R} = E_0 \cos \theta, \quad \varphi = -E_0 R \cos \theta$$

$$\text{边界条件} \quad \left\{ \begin{array}{l} \varphi_{\text{内}}|_{R=R_0} = \varphi_{\text{外}}|_{R=R_0} \\ \epsilon_s \frac{\partial \varphi_{\text{内}}}{\partial R} \Big|_{R=R_0} = \epsilon_0 \frac{\partial \varphi_{\text{外}}}{\partial R} \Big|_{R=R_0} \end{array} \right.$$

$$\begin{aligned} \Rightarrow \left\{ \begin{array}{l} \sum_l a_l R_0^l P_l(\cos \theta) = -E_0 R_0 \cos \theta + \sum_l \frac{b_l}{R_0^{l+1}} P_l(\cos \theta) \\ \epsilon_s \sum_l l a_l R_0^{l-1} P_l(\cos \theta) = \epsilon_0 [-E_0 \cos \theta - \sum_l (l+1) \frac{b_l}{R_0^{l+2}} P_l(\cos \theta)] \end{array} \right. \\ \left\{ \begin{array}{l} a_l R_0 = \frac{b_l}{R_0^{l+1}} - E_0 R_0 \\ \frac{\epsilon_s}{\epsilon_0} a_l = -\frac{2b_l}{R_0^{l+1}} - E_0 \end{array} \right\} \quad \left\{ \begin{array}{l} a_l R_0^l = \frac{b_l}{R_0^{l+1}} \\ \frac{\epsilon_s}{\epsilon_0} a_l R_0^{l+1} = -\frac{(l+1)b_l}{R_0^{l+1}} \quad (l \neq 1) \end{array} \right. \end{aligned}$$

$$\text{解得} \quad b_1 = \frac{\epsilon_s - \epsilon_0}{\epsilon_s + 2\epsilon_0} E_0 R_0^3, \quad a_1 = -\frac{3E_0}{\epsilon_s + 2\epsilon_0} E_0$$

$$b_l = a_l = 0 \quad (l \neq 1)$$

$$\left\{ \begin{array}{l} \varphi_{\text{内}} = -\frac{3E_0}{\epsilon_s + 2\epsilon_0} E_0 R \cos \theta \\ \varphi_{\text{外}} = -E_0 R \cos \theta + \frac{\epsilon_s - \epsilon_0}{\epsilon_s + 2\epsilon_0} E_0 \frac{R_0^3}{R^2} \end{array} \right.$$

讨论:

1)

$$\text{介质球中: } \vec{E}_{\text{内}} = -\nabla \varphi = \frac{3E_0}{\epsilon_s + 2\epsilon_0} \vec{E}_0$$

$$\text{退极化场: } \vec{E}' = \vec{E}_{\text{内}} - \vec{E}_0 = \frac{\epsilon_0 - \epsilon_s}{\epsilon_s + 2\epsilon_0} \vec{E}_0$$

$$\text{极化强度: } \vec{P} = \chi \vec{E}_{\text{内}} = (\epsilon_s - \epsilon_0) \cdot \frac{3E_0}{\epsilon_s + 2\epsilon_0} \vec{E}_0$$

$$\text{总电偶极矩: } \vec{p} = \frac{4\pi}{3} \lambda R_0^3 \vec{P} = 4\pi \epsilon_0 R_0^3 \cdot \frac{\epsilon_s - \epsilon_0}{\epsilon_s + 2\epsilon_0} \vec{E}_0$$

$$\varphi_P = \frac{\vec{p} \cdot \vec{R}}{4\pi \epsilon_0 R^3} = \frac{\epsilon_s - \epsilon_0}{\epsilon_s + 2\epsilon_0} E_0 \frac{R_0^3}{R^2}$$

$$\Rightarrow \varphi_{\text{外}} = \varphi_{\text{原}} + \varphi_P$$

$$2) \chi_r = \frac{\vec{P}}{\epsilon_0 \vec{E}_0} = 4\pi R_0^3 \frac{\epsilon_s - \epsilon_0}{\epsilon_s + 2\epsilon_0}$$

对金属纳米颗粒, 在特定波长 λ_r , 频率 ω_r

$$\epsilon_s(\omega_r) + 2\epsilon_0 \approx 0$$

称为局域等离子体共振.

4. 柱形边界

$$\nabla^2 \varphi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \phi^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

$$\varphi = R(r) G(\phi) Z(z)$$

$$\Rightarrow \begin{cases} r \frac{d}{dr} \left(r \frac{dR}{dr} \right) + (Kr^2 - \nu^2) R = 0 \\ \frac{d^2 G}{d\phi^2} + \nu^2 G = 0 \\ \frac{d^2 Z}{dz^2} - K^2 Z = 0 \end{cases}$$

$$\varphi(r, \phi, z) = \sum_{\nu, K} R_{\nu, K}(r) G_{\nu}(\phi) Z_K(z)$$

5. 柱形边界二维边值问题.

$$\nabla^2 \varphi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \phi^2} = 0$$

$$\varphi = R(r) G(\phi)$$

$$\hookrightarrow \begin{cases} r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} = \nu^2 R \\ \frac{d^2 G}{d\phi^2} + \nu^2 G = 0 \end{cases}$$

$$\text{通解 } \varphi(r, \phi) = (A_0 + B_0 \ln r) (\cos 0 + \sin 0 \phi) + \sum_{\nu} (A_{\nu} r^{\nu} + B_{\nu} r^{-\nu}) [C_{\nu} \cos(\nu \phi) + D_{\nu} \sin(\nu \phi)]$$

例: 将一介电常数为 ϵ , 半径为 R_0 的无限长圆柱形介质线放置于均匀电场 $\vec{E}_0$ 中

求介质线内的电场分布.

$$\varphi_{in}(r, \theta) = \sum_{n=1}^{\infty} [a_n \cos n\theta + b_n \sin n\theta] r^n \quad (r < R_0)$$

$$\varphi_{out}(r, \theta) = -E_0 r \cos \theta + \sum_{n=1}^{\infty} [c_n \cos n\theta + d_n \sin n\theta] \frac{1}{r^n} \quad (r > R_0)$$

$$1. \varphi_{in}|_{r=R_0} = \varphi_{out}|_{r=R_0}$$

$$\epsilon \frac{\partial \varphi_{\text{int}}}{\partial r} \Big|_{r=R_0} = \epsilon_0 \frac{\partial \varphi_{\text{ext}}}{\partial r} \Big|_{r=R_0}$$

$$\hookrightarrow \sum_l (a_l \cos l\phi + b_l \sin l\phi) R_0^l = -\epsilon_0 R_0 \cos\phi + \sum_l (c_l \cos l\phi + d_l \sin l\phi)$$

$$\frac{\epsilon}{\epsilon_0} \sum_l (a_l \cos l\phi + b_l \sin l\phi) (R_0^{l-1}) = -\epsilon_0 \cos\phi - \sum_l (c_l \cos l\phi + d_l \sin l\phi) \frac{1}{R_0^{l+1}}$$

$$\left\{ \begin{array}{l} a_l R_0 = -\epsilon_0 R_0 + \frac{c_l}{R_0} \\ \frac{c_l}{R_0^l} = a_l R_0^l \quad l \geq 2 \\ \frac{d_l}{R_0^l} = b_l R_0^l \quad l \geq 1 \end{array} \right. \quad \left\{ \begin{array}{l} \epsilon a_1 = -\epsilon_0 (\epsilon_0 + \frac{c_1}{R_0^2}) \\ \epsilon a_l l R_0^{l-1} = -\epsilon_0 c_l \frac{1}{R_0^{l+1}} \\ \epsilon b_l l R_0^{l-1} = -\epsilon_0 d_l \frac{1}{R_0^{l+1}} \end{array} \right.$$

$$\text{mit} \quad a_1 = -\frac{2\epsilon_0}{\epsilon + \epsilon_0} \epsilon_0, \quad c_1 = \frac{\epsilon - \epsilon_0}{\epsilon + \epsilon_0} \epsilon_0 R_0^2$$

$$a_l = c_l = 0, \quad b_l = d_l = 0$$

$$\varphi_{\text{int}} = -\frac{2\epsilon_0}{\epsilon + \epsilon_0} \epsilon_0 r \cos\phi$$

$$\varphi_{\text{ext}} = -\epsilon_0 r \cos\phi + \frac{\epsilon - \epsilon_0}{\epsilon + \epsilon_0} \epsilon_0 \frac{R_0^2}{r} \cos\phi$$

$$\vec{E}_{\text{int}} = \frac{2\epsilon_0}{\epsilon + \epsilon_0} \vec{E}_0, \quad \vec{P} = \frac{2\epsilon_0(\epsilon - \epsilon_0)}{\epsilon + \epsilon_0} \vec{E}_0$$

$$\vec{P} = \chi_e R_0^2 \vec{P} = 2\chi_e R_0^2 \frac{\epsilon - \epsilon_0}{\epsilon + \epsilon_0} \vec{E}_0$$